BM26 Rear Toe-Link Calculations

Current design → Heim Bore-Bore length = { Min ≈ 11.2 in (20 engaged threads) ; Max = 12 in (6 engaged threads) } (tube length may be reduced to reach preffered range of length)

4130 steel tube length = 9.0 inches (Not including Hex inserts/Heims)

Hex insert thread Calculations

(Critical Bending)

$$\tau_x = \frac{6f}{\pi (d_r)(n_t)(P)}$$

f = force applied
d_r = root diameter
n_t = # of threads engaged
P = Pitch

assuming for worst case;

$$\tau_x = \frac{6\,(260\text{ lbs})}{\pi\,(0.159\text{ in})(6)(1/32\text{ in})}$$ → Max load (T/C)

$$\tau_x = 16700 \text{ Psi}$$

$$F.S = \frac{S_y}{\tau_x} = \frac{54000}{16700}$$ → low carbon steel S_y (source: Mucaster)

$$F.S = 3.2$$

(If 6+ threads engaged, or generic bolts 6 threads take ≥ 95% of the load)

tube buckling calculations

(Pin-Pin Connection) (Assuming Bore-Bore Max length = L = 12 in)

∴ C = 1, n = 1

Check for Eulers Criterion → $\frac{L}{K} \geq \sqrt{\left(\frac{2\pi^2 CE}{S_y}\right)}$

$$98.4 \geq \sqrt{\frac{2\cdot\pi^2\cdot(1)\cdot(29700\times10^3\text{ Psi})}{70000\text{ psi}}}$$

$$98.4 \geq 91.51$$

since true; ∴ consider as a long column

Critical Load for long column → $P_{cr} = \frac{\pi^2 CEA}{L^2}$

$$= \frac{\pi^2 (1)(29700\text{ ksi})\left(\frac{\pi}{4}\cdot(0.375^2 - 0.319^2)\right)}{12\text{ in}}$$

$$P_{cr} = 928 \text{ lbs}$$

Buckling F.S $= \frac{928}{260} = 3.57$